

? React Interview Questions

What are the features of React?

- React is used to build *SPAs (Single Page Applications)*
- Creates *reusable UI components*
- ReactJS uses a Virtual DOM mechanism to efficiently update the HTML DOM.
- Faster Navigation (No full page reloads)
- Allows writing HTML-like syntax inside JavaScript through JSX.
- Data flows in a single direction (parent → child).
- With React Native, the same concepts can be used to build mobile apps for Android and iOS.
- Huge ecosystem of libraries, tools, and community support.

What's the use of Babel?

Babel is a JavaScript transpiler, which is a specific type of compiler mainly used to convert modern JavaScript (ES6+ and beyond) code into backward-compatible JavaScript that can run in older browsers or environments.

- **JSX Support:** Converts JSX (used in React) into plain JavaScript that browsers can understand.

Difference between Virtual DOM & Real DOM

Feature	Real DOM	Virtual DOM
Definition	The actual DOM (Document Object Model) rendered by the browser.	A lightweight in-memory representation (copy) of the real DOM maintained by React.
Updates	Directly updates the UI.	Updates happen in the virtual DOM first, then React compares (diffing) and updates only the changed parts in the real DOM.
Performance	Slower when frequent updates occur (manipulating real DOM is expensive).	Faster updates due to efficient diffing and batching of changes.
Re-rendering	Entire UI or large parts may re-render even for small changes.	Only the changed nodes are re-rendered in the real DOM after comparison.
Memory Usage	Takes more memory and resources.	Lightweight, stored in memory only.
Example	<code>document.getElementById()</code> directly modifies elements.	React internally creates a virtual DOM tree to update UI efficiently.

What are the Features of ES6?

- Let and Const - Block-scoped variables. let for mutable values, const for immutable values.
- Arrow Functions - Shorter function syntax. Lexically binds this (no need for .bind()).
- Template Literals - Use backticks (`) for string interpolation. Supports multi-line strings.
- Default Parameters - Set default values for function parameters directly.
- Destructuring Assignment - Extract values from arrays/objects into variables easily.
- Spread and Rest Operators (...) - Spread: Expand arrays/objects.
- Promises: Built-in support for asynchronous operations. Replaces complex callback patterns.

Difference between Class Components & Functional Components in React?

Aspect	Class Components	Functional Components
Definition	ES6 classes that extend <code>React.Component</code> .	Plain JavaScript functions that return JSX.
Syntax	More verbose, use <code>class MyComponent extends React.Component</code> .	Simpler, just <code>function MyComponent() { return <JSX/> }</code> .
State Management (Before Hooks)	Can hold and manage state using <code>this.state</code> and <code>this.setState()</code> .	Initially stateless (until Hooks were introduced).
Lifecycle Methods	Have built-in lifecycle methods like <code>componentDidMount</code> , <code>componentDidUpdate</code> , etc.	Lifecycle handled using Hooks like <code>useEffect()</code> .
Hooks Support	Not needed (state & lifecycle already built-in).	Use Hooks (<code>useState</code> , <code>useEffect</code> , etc.) for state & lifecycle features.
Performance	Slightly heavier due to <code>this</code> binding and more boilerplate.	Lighter and faster because they're just functions.
'this' Keyword	Must bind <code>this</code> to event handlers manually or use arrow functions.	No <code>this</code> keyword, easier to read & maintain.
Recommended Usage	Older React apps (before Hooks).	Modern React apps (preferred approach today).

What's `useState`

`useState` is a React Hook that allows you to add state to functional components.

Syntax:

```
const [state, setState] = useState(initialValue);
```

`state` → current state value.

`setState` → function to update the state.

`initialValue` → starting value of the state.

Key Points:

- State is preserved between re-renders.
- Updating state with `setState` causes the component to re-render.
- You can store any type (string, number, array, object, etc.).
- State updates are asynchronous and batched.

What's the difference between Controlled & Uncontrolled Components in React?

Aspect	Controlled Component	Uncontrolled Component
Definition	Form data is handled by React state .	Form data is handled by the DOM itself.
Data Source	React state is the single source of truth.	The DOM (via refs) is the source of truth.
Value Prop	Input value is bound to a state variable (<code>value={state}</code>).	Input value is not bound to state; accessed only when needed.
Change Handling	Every change triggers an <code>onChange</code> event to update state.	No <code>onChange</code> needed to update state continuously.
Accessing Value	Directly from React state.	Using <code>ref</code> or <code>document.getElementById()</code> to read value.
Example Usage	When you need validation, conditional disabling, or real-time updates.	When you just need the final value at submission time.

What's Prop Drilling?

Prop drilling happens when you pass data (props) from a parent component to a deeply nested child component by passing it through multiple intermediate components that don't actually need the data.

Why It Happens:

React only allows data to flow one way (top → down), so if a deeply nested component needs data, you often have to pass it through every component in the hierarchy.

Problem:

- Intermediate components become cluttered with props they don't use.
- Code becomes harder to maintain.

- Changes to props can cause unnecessary re-renders.

Solutions to Avoid Prop Drilling:

- React Context API (most common)
- State Management Libraries (Redux, Zustand, Recoil, Jotai)
- Custom Hooks (if only logic is shared)